

Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

Supplementary Appendix

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Table S1 Dimensions and Drivers of Food and Nutrition (In)security

Dimension	Description	Drivers
Availability	Enough safe, healthy, nutritious, diverse, and culturally acceptable food is available in the communities in which people live, either through domestic production or imports.	Food availability is affected by agriculture, transportation, manufacturing, international trade, and social policies. The largest global threats to food availability are climate change, political conflict, and economic underdevelopment. In low-income countries, nutritious food availability is limited due to the unequal distribution of food globally, lack of agricultural infrastructure and trade agreements that increase availability of nutrient-poor foods. In high-income countries, there is enough food for most of the population, but marginalized, lower income, and geographically-isolated populations have issues accessing affordable nutrient-dense foods. ²
Access	Physical access: there is sufficient infrastructure and means for people to physically procure foods without undue burden on household resources. Economic access: there is sufficient income to meet the needs of daily living and food is affordable. Social access: food can be accessed irrespective of personal circumstances or characteristics and its access does not require stigmatized behavior (e.g., begging).	The proportion of people living in urban areas is expected to increase from 55% to 68% by 2050. ³ The urbanization of populations reduces the agricultural workforce, of which women make up the largest proportion globally. Populations living in urban areas have less access to food and must rely on employment to purchase food. Being in good health, having access to quality education, living in affordable safe housing, affordable energy sources (for cooking, heating, cooling, and transporting food) all contribute to the ability to maintain economic stability.
Utilization	This dimension includes the body's effective absorption and metabolism of nutrients; access to reliable information on food safety; nutrition and food literacy; access to equipment and space to store, prepare, and eat food; and the intrahousehold distribution of food.	The body's effective uptake of nutrients requires access to clean water and food. Diarrheal disease caused by contaminated water and food is the most common cause of mortality for children under the age of five. ⁴ Preventing diarrheal disease requires knowledge of and infrastructure for food sanitation. Trade-offs affect how food is distributed within a household; for example, male household members engaged in physical labor may preferentially receive food ⁵ or children in high-income countries may preferentially receive food compared with adults, especially female caregivers. ⁶ Converting ingredients to palatable and bioavailable food requires food literacy, access to equipment, and fuel for cooking. It also requires the time necessary to plan, purchase, and prepare food.

Stability	The availability of, access to, and utilization of nutritious food is embedded in a stable system that ensures food irrespective of seasonal variation, weather events, natural disasters, pandemics, violence (civil and domestic), and economic volatility.
Sustainability	The stability dimension accounts for shorter-term disruptions, while the sustainability dimension focuses on the connections between ecosystems and economies to support food security across generations. Food practices, economies, and social systems need to contribute to the long-term regeneration of resources. Food systems produce the most greenhouse gas emissions, and moving toward more sustainable food systems will be essential to ensure future food security.
Agency	Agency involves the capacity for persons and communities to make choices about food and how it is accessed. Food should be acquired in ways that are socially acceptable and just. Agency is affected by power imbalances between countries, their citizens, and corporations, as well as differential access to resources and information. Agency implies that persons and communities can create a food system that meets their needs and is environmentally, economically, and socio-culturally sustainable.

Table S2: Global Food Security, 2024

In 2024 the world is experiencing increasing rates of severe food and nutrition insecurity.

Some of the contributing drivers to this increase include:

- In Africa an estimated 75 million people will be FI in 2024 due to extreme weather events (drought, flooding), deteriorating economic conditions, increasing cereal (wheat and maize) prices and ongoing conflict. ⁷
- In Asia and the Pacific climate shocks will decrease crop yields and while there has been a recent decline in food prices, rice price remains high due to low yields and export restrictions.
- The Israeli Palestinian conflict has limited aid to reaching the Gaza Strip, with food being used as a weapon. ⁸. In Lebanon, Israeli strikes have led to 75 per cent of farmers losing their primary source of income. Across the Middle East food prices have increased due to rises in transportation and agricultural inputs. ⁷
- In Latin America, Argentina is hardest hit with rising wheat flour prices and an annual food inflation rate of 300 per cent aggravated by currency devaluation. Political instability, natural disasters and economic uncertainty has resulted in an emergency situation in Haiti, where 22 per cent of children under the age of five are chronically malnourished. ⁷
- The COVID pandemic and ensuing global lockdown highlighted the fragility of the food supply chain at global, national and local levels and the urgent need for food system transformation. The pandemic has resulted in a global economic recession resulting in loss of incomes and livelihoods, widening social inequalities, disruptions to social protection programs and food price volatility. ⁹
- The FAO Food Price Index has increased from between 25-50% since 2020, peaking in 2022 and slowing decreasing in 2024. However, in the first four months of 2024 remains 25% above the pre-pandemic (2019) index. ¹⁰
- The global economic recession is contributing to food inflation and this in combination with currency depreciation increasing debt and further increasing food, energy and agricultural input prices increasing the number of people experiencing hunger.
- Russia and Ukraine account for over 15%, 25% , and 60% of world maize, wheat, and sunflower oil exports. ¹¹ Conflict in this area has disrupted production and supply chains combined with trade restrictions has increased cereal prices. The region is main supplier of fertilizer and export restrictions have seen prices escalate and the increase cost of agricultural inputs will drive food prices higher. ¹¹ Additionally, conflict here and in the Middle East has raised fuel prices globally contributing to household trade-offs regarding transport and household energy utilisation.
- Climate change is reducing biodiversity, increasing the risk of plant disease and weather changes will impact on the ability to grow enough food to feed ever-growing populations. This makes addressing post-harvest food waste imperative. ¹²
- Extreme weather events have been identified as the key climate change risk affecting the planet, extreme heat, precipitation and drought have increased in nearly all regions of the world. These extremes have negatively affected food production and water availability in most regions of the world but particularly in Africa and small island nations states. ¹³

Table S3: Selection of Tools Used to Measure Food and Nutrition Insecurity

Tool	What is measured?			Description	Advantages and disadvantages
	H	FI	NI		
Prevalence of Undernourishment (PoU) indicator	✓			Used by FAO as an indicator to determine progress on Sustainable Development Goal 2 – “Zero Hunger”. It is an estimate of dietary energy adequacy. Measures percentage of a population who are not meeting food requirements to meet daily dietary energy requirements. FAO defines hunger as being synonymous with chronic under nourishment.	Aggregate indicator drawing on average availability of calories, variation in energy intake within a population and demographic data to derive a minimum energy requirement. Is calculated using statistical modelling. Does not provide data on households or individuals. Does not consider diet quality or diversity. A macro-indicator to assess global inequities in access to enough food.
Global Food Insecurity Index (GFSI)		✓	✓	The GFSI is a composite index of 25 indicators providing an aggregate score (0-100). Covers the four dimensions of FI including availability, access, utilisation and stability.	High level indicators do not provide information at household or personal level. Some indicators of nutrition security. Is valid and dependable. Does not include outcome indicators.
FAO – Suite of Food Security Indicators		✓	✓	This suite of indicators provides high level national indications of food security across availability (of dietary energy and protein), access (physical infrastructure, economic access), stability (including political stability), utilization (drinking water, sanitation, child anthropometrics, adult obesity, exclusive	High level indicators do not provide information at household or personal level. Some indicators of nutrition security. Includes outcome indicators. Few indicators of diet quality/diversity. Reliant on accurate, timely national data.

				breastfeeding, birthweight).	
Anthropometric measurements Prevalence of micronutrient deficiencies		✓	✓	May be used to diagnose malnutrition, growth faltering, underweight, micronutrient deficiencies (particularly iron, iodine), overweight, and obesity in adults and children.	Outcome indicators of food and nutrition security, specifically of the utilization dimension. Reflects the “active and healthy life” component of the definition.
FAO Cost of a Healthy Diet (CoHA)		✓	✓	Affordability of a healthy (diversity, adequacy, moderation and balance) based on national food-based dietary guidelines, using the cheapest available local foods compared with national incomes after considering cost of essentials e.g., housing.	Provides national level data using data that is collected by most countries to monitor inflation. Does not drill down to at risk groups within-country. Does not provide data at household or personal level.
Food insecurity experience scales	✓	✓		For example, scales such as Household Food Insecurity Access Scale (HFIAS), Household Food Security Survey Module (HFSSM), Latin American and Caribbean Household Food Insecurity Scale (ESCLA) and the Food Insecurity Experience Scale (FIES). Can provide an indication over 4 weeks, three months up to 12 months. The FIES was developed by the FAO to allow comparison across countries and is used in conjunction with PoU as an indicator to determine progress of the “zero hunger” Sustainable Development Goal.	Measures financial access at the household level. Does not capture procuring foods through hunting, gathering or gardening. Gives an indication of severity at the household level. Does not take into consideration intrahousehold distribution of food or nutrition security. Underestimates those who are able to afford food but may be limited in their ability to acquire foods that meet, health needs and cultural preferences. These tools are valid and dependable and have been used across multiple country contexts. They do make assumptions about what constitutes a “household” and may exclude those who are

					homeless or who live in situations that are more collectivist and there is a fluidity of resources across physical structures. These tools are valid and dependable and have been used across multiple country contexts. They have not been used in First Nations communities.
Dietary diversity measures such as the Household Dietary Diversity Scale (HDSS)		✓		Availability of between 12-16 nutrient-dense foods in a household. Measures availability only and not consumption.	Measures household availability of diverse foods. Is an indicator of diet quality but does not measure caloric adequacy. Not valid or dependable.

Note: H=Hunger; FI= Food insecurity NI= Nutrition insecurity¹⁴⁻¹⁷

Table S4: Severity of Food Insecurity as Measured by Food Insecurity Experience Scales

Food Security	Mild/Marginal Food Insecurity	Moderate Food Insecurity	Severe Food Insecurity
There are no indications of problems with food access or insufficient diet quality or quantity.	There may be anxiety over food insufficiency or shortages. There are few or no indications of insufficient food quality or quantity.	There is reduced diet quality or variety. This may include consumption of energy-dense, nutrient-poor foods or a diet lacking variety. There is few or no indications of reduced food intake.	There is evidence of disrupted eating patterns (e.g. skipping meals) and reduced food intake. Affected persons are hungry but not able to eat or for a day or multiple days.

Severity of Food Insecurity^{18, 19}

Table S5: Households and Individuals at Higher Risk of Food and Nutrition Insecurity

Demographic, household and equity factors	Women²⁰ Pregnant and lactating women²¹ Children²² Younger people²³; University/college students²⁴ Older adults due to functional and financial limitations.²⁵ Older adults with financial stressors such as being in debt, not owning their own home were at higher risk²⁶ Single persons, living on their own^{23 27} Men living alone^{20, 23, 27} Large households^{28, 29} Households with children^{2, 30-32} Women-headed households^{2, 20, 33} and single-parent households² Grandparent-headed households^{34, 35} LGBTQIA+³⁶ People of colour^{30, 33} Migrants and refugees³⁷⁻³⁹ First Nations⁴⁰⁻⁴² Military veterans^{43, 44} People with a history of incarceration or on probation or parole⁴⁵
Income, education and housing	Poverty^{33, 46 27, 41, 47, 48} Unemployment; precarious employment, low paid employment: (those without guaranteed hours or employee rights such as minimum wage and sick leave)^{2, 30, 48} Reliance on social insurance payments (e.g., unemployment insurance, single-parent cash transfers) in high-income countries (in low-income countries cash transfers can be protective against FI).^{31, 47, 49-51} Older adults in some countries are at lower risk of FI primarily due to the aged pension being indexed to inflation^{30, 46} Low educational attainment^{2, 30, 41} University/college attendance, limiting income and raising expenditure²⁴ Housing insecurity: homelessness, precarious housing, housing hardship, renters^{52, 53 54-56} Rural households² Urban households in low and middle-income countries⁵⁷⁻⁵⁹
Health	Living with a disability⁶⁰⁻⁶² Living with a serious mental illness,^{62, 63} or disorders of executive functioning (e.g. ADHD, ASD).⁶²⁶⁴ Domestic, family and intimate partner violence, adverse childhood experiences, racism and discrimination^{65 66-68 69}; Living with non-communicable diseases^{70 63} Acute infectious disease e.g., COVID and infectious parasitic diseases⁷¹,

Emergencies	External to the household: political conflict, natural disasters, pandemics⁷²⁻⁷⁵ Internal to the household: job loss, death of a spouse or child, loss of relationships, intimate partner and domestic violence⁶⁵
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Table S6: Health Consequences Associated with Food Insecurity

Associations have been listed that control for other indicators of poverty including income, education, socio-demographic characteristics (sex, age, race etc.) and other relevant health factors. However, it is unlikely that controlling for these indicators establishes a causal pathway between food insecurity and health, which is independent of the array of structural and social drivers and ramifications of poverty. When diet diversity and diet quality have been considered reference is made to food and nutrition insecurity (FNI), otherwise the association is with food insecurity (FI)

Pregnant women and children	
Gestational weight gain and pregnancy outcomes	Low-income countries - FNI is associated with micronutrient deficiencies, low dietary diversity and low caloric intake which can lead to inadequate weight gain, preterm birth, and low birthweights.⁷⁶ Pre-term birth is the leading cause of death in newborns.⁷⁷ Middle and high-income countries - Pregnant women in households with mild and moderate FI are at risk of weight gain above international recommendations.^{78, 79} - After accounting for pre-pregnancy BMI, demographic and socioeconomic factors, there is evidence that pregnant women who are living in FI households gain significantly less weight than their FS counterparts.⁷⁸ - Pregnant women in households with severe FI are at risk of inadequate gestational weight gains.⁷⁹ - Pregnant women experiencing FI may be at higher risk of GDM but the association could be partly explained or compounded by weight status and GWG.^{80, 81} - FI pregnant women are not at risk of complications due to gestational hypertension or diabetes compared with FS women.⁸² - There is limited data on pregnancy outcomes but population wide data from Canada suggests there are no associations between FI and premature births or infants born small or large for gestational age potentially due to country-specific supports for at-risk women.⁸²
Mortality	Global Nearly half of all deaths in children under five are linked to undernutrition which increases the mortality of infectious diseases, the largest contributor to neonatal and child deaths globally.⁷⁷
Weight status Wasting/Stunting	Low-income countries 22% of children under the age of five were stunted (short for age) seriously impacting their potential for cognitive development, 7% were wasted (thin for height), and 5.6% were overweight – increasing their risk of NCDs.⁸³

	• Micronutrient deficiencies in particular of iron, Vitamin A and zinc, linked to FNI, are limiting the growth and cognitive potential of children in LIC.⁸⁴ High-income countries • For infants living in FI households, FI does not appear to be associated with height or growth faltering but are potentially at risk of overweight. Based on MUAC, children in FI households are more likely to be underweight but by the age of five these children are more likely to be obese. There are no clear patterns between FI and weight status among children in HIC. There are no clear indications of dietary quality or variety.⁸⁵
Infectious disease	Low-income countries • Children experiencing FI who are undernourished are more susceptible to and have higher mortality from pneumonia,⁸⁶ and diarrhoeal disease.⁸⁷ High income countries There is evidence that hospitalization rates related to infections are higher for children from FI households.^{88, 89}
Child development	Global • All levels of FI severity are associated with risk of developmental delay, poor cognitive and academic outcomes, internalizing and externalizing behaviours and poor development of interpersonal skills.^{6, 90} • All levels of FI severity are associated with lower school readiness (in the domains of self-regulation and social-emotional development) and the need for increased support at school.⁹¹
Mental health and psychological distress	Global Adolescents with FI (aged 12 and above) are at higher risk of suicidal ideation, mood and anxiety disorders, psychological distress, and loneliness⁹²⁻⁹⁷
Disordered eating	• High income countries FI in adolescence is associated with unhealthy weight control behaviours (fasting, volitional vomiting, skipping meals, laxative use, diet pill use, diuretic use, smoking more cigarettes).⁹⁸ • Adolescents living in FI households display a greater frequency of binge eating independent of other factors.⁹⁹ • A history of HFI as an adolescent contributes to unhealthy weight control behaviours, binge eating disorder and elevated BMI as a young adult.⁹⁸ • Distorted eating is associated with cultural norms around food distribution where boys may be preferentially fed over girls^{100, 101} in LIC and MIC. In HIC children are often protected from hunger by adults for as long as possible.⁶ •

Adults	
Mortality Reduced life expectancy	Global • FI and especially severe FI is associated with premature death, all-cause (except for cancer) mortality^{71, 102, 103}. • Diet is the third largest contributor to the global burden of disease and in all countries NCDs are emerging as the leading cause indicating FNI.¹⁰⁴ • In sub-Saharan Africa FNI is associated with reduced life expectancy.¹⁰⁵ • There is emerging data that most countries in the world are now experiencing the double or triple burden of disease due to FNI, with increasing deaths related to NCDs.¹⁰⁶
General health	FI is associated with poor physical health, mental health and wellbeing and quality of life.^{97, 107-109} FI is associated with lower reports of wellbeing in HIC and MIC but not in LIC.¹¹⁰
Infectious diseases	Low-income countries • FI is associated with increased susceptibility to respiratory tract infections, diarrheal disease, human immunodeficiency virus (HIV) and tuberculosis.¹¹¹⁻¹¹³. • Specific to tuberculosis FI is associated with progression from latent to active disease as well as co-infection with HIV.^{111, 112, 114, 115}
Mental health	Global • FI is associated with mental in men and illness including anxiety disorders, mood disorders (depression, bipolar disorder mania or dysthymia), psychological distress, depression and suicidal ideation; as well as stress and sleep disorders.¹¹⁵⁻¹²¹ • A review of association of FI with mental illness in Africa revealed a dose-dependent impact especially for pregnant women and people living with HIV.¹²²
Weight status	Low-income countries • Severe FNI is associated with underweight among the poorest women¹²³ Middle and high-income countries • FI of all severities is associated with increased risk of overweight and obesity in adults, with women more susceptible than men.¹²⁴ • Severe FI could also be associated with underweight among women.^{124, 125}
Non-communicable diseases	Global • An association between non-communicable diseases and FI in low-income countries is emerging. For example, in sub-Saharan Africa FI is associate with a 41.8% prevalence of metabolic risk factors (e.g obesity, hyperlipidaemia, hypertension).¹²⁶ • The impact of poor in-utero nutrition and low birth weights and access to energy-dense processed foods is

	increasing and contributing to higher NCD rates in LIC and MIC ¹²⁷. High income countries • There is a dose-dependent association between FI and metabolic syndrome, ¹²⁸ coronary artery disease risk, ¹²⁹⁻¹³¹ diabetes and asthma, ¹²⁹ hypertension. ^{131, 132} • FI is also associated with chronic pain, ¹³³ restrictive lung disease, ¹³⁴ asthma, ¹³⁵⁻¹³⁸ dental caries, ¹³⁹ kidney stones, ¹⁴⁰ metabolic dysfunction associated liver disease. ¹⁴¹
Diabetes	High income countries • Adults with FNI and T2DM were more likely to have poor glycaemic control (HbA1c $\geq 7.0\%$). ¹⁴² Those with poor diets and FI have significantly higher odds of poor glycaemic control. ^{143, 144} • FI is associated with missing eye examinations and vision loss, ^{144, 145} diabetes medication non-adherence, ¹⁴⁶ and being hospitalized, ¹⁴⁷ •
Disordered eating	• High income countries FI is associated with disordered eating including emotional eating, ¹⁴⁸ dietary restraint, loss of control over-eating, ¹⁴⁹ and unhealthy weight behaviours such as laxative use. ⁹⁹ • Moderate and severe FI is associated with binge eating disorder and bulimia nervosa. ¹⁵⁰⁻¹⁵²
Older adults	
General health	FI is associated with cognitive impairment, and faster declines in executive function. There are also associations with sarcopenia, increased risk of falls and frailty. ^{115, 153-160}

Legend - ADHD: Attention Deficit Hyperactivity Disorder; BMI: Body Mass Index; COPD: Chronic Obstructive Pulmonary Disease; CVD: Cardiovascular Disease; FI: Food insecure/Food insecurity; FS: Food secure/Food security; GDM: Gestational Diabetes Mellitus; GWG: Gestational Weight Gain; LFS: Low Food Security (Moderate Food Insecurity); VLFS: Very Low Food Security (Severe Food Insecurity); HFI: Household Food Insecurity; MUAC: Mid Upper-Arm Circumference; NAFLD: Non-alcoholic Fatty Liver Disease; T2DM: Type 2 Diabetes Mellitus

Table S7: Nutrition-specific and Nutrition-sensitive Interventions for Food and Nutrition Security

Nutrition-specific Interventions	Nutrition-sensitive Interventions
Climate change mitigation, reversal of de-forestation, securing water supplies, enhancing biodiversity, improving yields, reducing greenhouse gas emissions.^{12, 161} As the threat of conflicts, natural disasters and economic shocks increase there are calls for countries to develop food security crisis preparedness plans. These plans rely on accurate data to enable governments to develop cohesive operational frameworks to identify and mitigate risks and to develop timely actions that will prevent severe FI during emergencies.^{162, 163}	Upholding human, child, Indigenous rights including the right to food.¹⁶⁴
Food system transformation that builds agency including application of agroecological and food sovereignty principles.¹⁶⁵ Food systems need to be resilient to shocks to retain stability over time.^{75, 166}	Nutrition sensitive agriculture interventions via four paths: improved availability of and access to nutrient-rich foods, agricultural income increasing purchasing power, agricultural policies, impacting food prices, agricultural interventions providing opportunities for women and their children.¹⁶⁷ Interventions include homesteading, home gardens, irrigation projects, biofortification.¹⁶⁸
Building equitable and sustainable food supply and value chains, which ensure food safety and include waste reduction across the food system.¹²	Ensuring people have access to resources that ensure that can generate an income, including access to loans, productive assets (e.g., tools), employment, training and support¹²
Stabilizing food prices.¹⁶⁹	Providing social protection to ensure those with limited earning capacity are adequately financially supported.^{31, 46, 50, 170, 171} Social protection and income support that provides a living wage to afford adequate shelter, food, health and other necessities. Social assistance (cash transfers, in some context cash transfers + food, is more effective) and social insurance (unemployment, age, disability, child benefits) that protects households from shocks that could result in loss of income or in lower income earning potential. Cash benefits for families (income support for families, birth or maternity grants, financial support for low-income families with children, income support for childcare and

	school) reduces the risk of FI in those countries with such policies.³¹ The COVID pandemic provided a natural experiment, where, in most jurisdictions, social welfare payments and low wages were supplemented resulting in decreased reliance on food relief.¹⁷² Unconditional and conditional cash transfers may improve some health outcomes (including food security and diet diversity), and may improve school attendance and healthcare expenditure.^{51, 173, 174}
Promoting breastfeeding and counteracting the marketing of artificial milks for infants and children.^{175,176}	Economic growth is not enough to ensure food security equality with actions necessary to build household income. Ensuring a minimum wage, wages growth and employee benefits (e.g., sick leave, guaranteed hours) are essential as is the indexing of social insurance to CPI increases to ensure that income inequities are not exacerbated.⁴⁸
Fortifying food. Food fortification, bio-fortification and supplementation. Food fortification programs include the deliberate addition of micronutrients to foods to ameliorate micronutrient deficiencies. The most commonly implemented fortification programs include the addition of iodine to salt, folate to wheat products, and iron to a range of foods^{177, 178}. Biofortification refers to either traditional breeding or transgenic approaches to increase the micronutrients and their availability in commonly consumed plants, an example would be biofortification of bananas with Vitamin A.¹⁷⁹ Supplementary feeding is often provided to FI individuals to increase the nutritional value and adequacy of the diet, it is used in emergency and non-emergency situations. These can be additional foods to the normal diet or specially formulated foods (such as ready-to-use therapeutic foods (RUTF)) or micronutrient supplements. The effect of supplementation on growth in children is mixed and there does not appear to be long-term benefits for adults who are HIV positive.¹⁸⁰	Providing universal access to affordable quality education.²

Improved access to food production resources – this is gender-based with women having less access to agricultural landholdings. ¹²	Universal access to affordable healthcare. When healthcare (treatment, prevention, medications) is costly, households may trade off access to services for food. ^{161,181}
Promoting equitable trade policies. International trade - Equitable liberal trade policies that take public health into consideration. While international trade and trade liberalisation has in some cases improved economic growth and the import of foods to increase diversity, however an analysis of liberal trade policy across 132 countries shows that it is associated with lower FI in HIC but with increased FI among the poorest households in LIC. ¹⁸²	Providing affordable, safe shelter (housing) – a major household expense is housing – and there is evidence that the provision of affordable housing directly decreases FI. ^{55, 183}
Strengthening of food regulatory and surveillance systems to maintain food safety. ¹⁸⁴	Providing affordable childcare. Female-headed households with children are at greater risk of FI, consequently, to engage in a livelihood and build earning capacity young children need to be cared for. A social network assists with this as does affordable childcare. ^{31, 185}
Ensuring energy security and affordability – the recent energy crisis has exacerbated households trading between heating their homes and eating; including access to clean fuel for cooking. ¹⁸⁶	Addressing commercial determinants of health including trade policies that promote importation and aggressive marketing of ultra-processed foods. ¹⁷⁶ Commercial determinants include for example, speculative trading in basic necessities such as food leading to food price volatility; trade and investment liberalisation that can increase economic growth but also access to health-harming foods (e.g., sugar sweetened beverages); privatisation of essential services (education, health, utilities) that has increased costs and reduced equitable access; corporate changes to settings to maximise sales making them obesogenic. ¹⁸⁷
Ensuring a safe, stable water supply (water security). Improving sanitation facilities. ^{188, 189}	Reducing inequities, structural racism, mitigating intimate partner and family violence – these have been described as transformative social protection measures. ⁶⁹
Providing access to food safety-nets in the form of food aid (including RUTF), nutrition sensitive food banks and pantries, food vouchers. In the USA this food and nutrition safety-net is provided to those on low incomes in the form of SNAP and WIC. ¹⁹⁰ There is strong international evidence	

indicating that using charitable food relief (food pantries and foodbanks) creates stigma and shame and contributes to poor mental health. ¹⁹¹	
Increasing access to nutritious food within health services using a food as medicine approach via co-located pantries, and medically tailored food boxes or prescriptions. ¹⁹⁰	
School meals – provide at least one meal a day for children has reduced wasting and improved diet quality promoted attendance, and enhanced learning (these are considered as an in-kind transfer to households) ¹⁹²	
Food-retail planning that optimises healthy food accessibility, eliminates food deserts and improves agency.	
Building food and nutrition literacy. Being able to convert food into meals with bioavailable nutrients requires technical knowledge and skills and is included in the utilisation domain. These can be implemented at a broader level or provided as personal food and nutrition education. ^{23, 193}	

Table S8: Hunger Vital Sign™ FI Screening Questions

1. Within the past 12 months, we worried whether our food would run out before we got money to buy more.
2. Within the past 12 months, the food we bought just didn't last and we didn't have money to get more.

Response options:

- often true
- sometimes true
- never true
- don't know/refused

A positive screen is answering 'often true' or 'sometimes true' to either question¹

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